# Affirmed Networks Documentation Library

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**Affirmed Networks Documentation Library**
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# Preface

This document lists and describes each document in the Affirmed Networks library.

## Audience

This document is intended for network operators (sometimes referred to as Users or Operators in this document), installation staff, and other qualified service personnel who want to install, provision, monitor and learn about the Affirmed Networks system and associated features and functions. The audience should also be familiar with Long Term Evolution (LTE), and 3G or 4G networks. It may also be helpful to have a basic understanding of the HTTP protocol, TCP/IP stack, and the use of MMS services.

## How to Obtain Product Documentation

Documentation is available for currently supported product releases. Documentation is available online in Adobe PDF format. You can view PDFs online using the Adobe Reader®. To download the latest version of the Adobe Reader software from the Adobe web site, go to http://get.adobe.com/reader/

Product documentation is available from the Affirmed Networks SFTP server. Contact your sales representative or the TAC for instructions on accessing the SFTP server. Refer to the release notes for the latest release version.

# Technical Assistance Center

Affirmed Networks Technical Assistance Center (TAC) provides technical support 24 hours a day, 7 days a week (24 x 7) for any customer with an active maintenance contract.

To contact the TAC:

E-mail: prodsupport@affirmednetworks.com

24 x 7 Telephone support:

Local: 1-978-268-0871

Toll-free: 1-888-283-2838

# Documentation Library

The following documents are available from Affirmed Networks.

| Title | Description |
|---|---|
| **Acuitas Service Management System** | |
| *Acuitas Performance Statistics* | Performance statistics viewable through the Acuitas Service Management System. |
| *Acuitas System Administration Guide* | Installation, configuration, operation and maintenance of the Acuitas Service Management System. |
| *Acuitas User's Guide* | Using the Acuitas Service Management System to perform tasks, such as adding network elements and using Topology, Configuration, Fault, Performance, Virtual Network Function Manager, Security, Tools, Workflow, and Configuration Navigator applications. |
| **Affirmed Service Automation Platform (ASAP)** | |
| *Affirmed Service Automation Platform Administration Guide* | ASAP platform installation and configuration, vBuilder and vTransactor software installation instructions, and ASAP server administration tasks, such as backup and restore, logging, and other ASAP database/server operations. |
| *Affirmed Service Automation Platform Operations and Maintenance Guide* | ASAP SNMP Management Information Base (MIB) trap objects and trap alarms, and the Host-Resources MIB supported by ASAP software. |
| *Affirmed Service Automation Platform vTransactor User's Guide* | ASAP vTransactor overview, concepts, and procedures for creating service instances from imported service models used to manage and roll out services (service orchestration) to devices in carrier networks. |
| *Affirmed Service Automation Platform vBuilder User's Guide* | ASAP vBuilder overview, concepts, and procedures for building service models as templates used to create and deploy service instances in carrier networks. |
| **Charging Gateway Function (CGF)** | |
| *Affirmed Networks Charging Gateway Function Administration Guide* | Installation and configuration of the Affirmed Networks Charging Gateway Function (CGF) as a standalone node and multi-node cluster. |

| Title | Description |
|---|---|
| **CIoT Serving Gateway Node (C-SGN)** | |
| *Affirmed Networks C-SGN Gateway Overview* | This guide describes an overview of the Affirmed Networks CIoT Serving Gateway Node (C-SGN) service running on the Mobile Content Cloud. This guide also describes the C-SGN gateway features and functions and the CLI objects and commands used to configure these features. |
| **Deep Packet Inspection (DPI)** | |
| *DPI Classification Release Notes for R8.4 and above* | Release notes for the applications and protocols supported by the DPI Signature in MCC R8.4 and later releases. |
| *DPI Classification Release Notes for R8.3 and earlier* | Release notes for the applications and protocols supported by the DPI Signature in MCC R8.3 and earlier releases. |
| *DPI Script User Guide* | This guide describes how Operators can enable DPI Script to add or update classification support for applications and protocols without requiring an MCC software update or restart. |
| **Enhanced Service Capability Exposure Function (eSCEF)** | |
| *Affirmed Networks Enhanced Service Capability Exposure Function Administration Guide* | Affirmed Networks eSCEF and its role in Cellular Internet of Things (IoT) and M2M communications, an overview of eSCEF, configuration information, statistics and eSCEF-generated KPIs. |
| *Affirmed Networks Enhanced Service Capability Exposure Function Application Programming Interface Reference* | Affirmed Networks eSCEF Application Programming Interface for the eSCEF 3GPP APIs. |
| *Affirmed Networks Enhanced Service Capability Exposure Function Deployment Guide* | Deploying the Affirmed Networks eSCEF, including its components (AN-SCEF VNF, DG VNF, and Redis Database VM). |
| *Affirmed Networks Enhanced Service Capability Exposure Function Portal User's Guide* | Affirmed Networks eSCEF Portal and the user interface used to configure and provide eSCEF services to customers in the Operator's IoT network. |
| *Affirmed Networks Enhanced Service Capability Exposure Function SNMP MIB and Alarm Reference Guide* | The process for displaying trap alarm information, the organization of the Affirmed Management Information Base (MIB), and information on accessing and reading the MIB. |
| *Affirmed Networks Enhanced Service Capability Exposure Function Troubleshooting Guide* | Information to identify and troubleshoot common issues associated with the Affirmed Networks eSCEF application. |

| Title | Description |
|---|---|
| **Interface Specifications** | |
| *Affirmed Networks Charging Data Records Specification* | Charging Data Records used for offline billing and accounting and HTTP transaction data records used to provide information about each HTTP transaction that traverses the Mobile Content Cloud HTTP Proxy. It also contains information on the offline charging interfaces (Ga and Gz) used to transport charging information from a gateway (SGW, PGW, GGSN) to an offline charging system. |
| *Affirmed Networks Gx Interface Specification* | Gx information used for carrying policy and charging control information between the Policy and Charging Enforcement Function (PCEF) and the Policy and Charging Rules Function (PCRF). |
| *Affirmed Networks Gy Interface Specification* | Gy interface used for carrying online-charging quota and data usage information between the GGSN/PGW and the Online Charging System (OCS). |
| *Affirmed Networks RADIUS Interface Specification* | RADIUS interfaces used for carrying authentication, authorization, and accounting information between the Affirmed Networks Mobile Content Cloud device and a centralized AAA (or RADIUS) server. |
| **Mediation Service Function (MSF)** | |
| *Affirmed Networks Mediation Service Function Administration Guide* | Installation and configuration of the Affirmed Networks Mediation Service Function (MSF) as a standalone MSF node and as a multi-node MSF cluster. It also provides information about the event data record (EDR) types, fields, and parameters. |
| *Affirmed Mediation Service Function Deployment Guide for RHEL 7.3 OpenStack Newton Using Packstack* | Installation and configuration of the Affirmed Networks Mediation Service Function (MSF) platform with SR-IOV using RedHat OpenStack Newton using Packstack. |
| **Mobile Content Cloud (MCC)** | |
| *Affirmed Active Intelligent vProbe System Administration Guide* | Affirmed Networks Active Intelligent vProbe and its features, configuration, and EDR record formats. |
| *Affirmed Networks CLI Reference Guide* | MCC Command Line Interface used to configure functions and system attributes. This Command Line Interface enables operators to enter commands to perform specific tasks to configure, control, query, and monitor system hardware and software components. |
| *Affirmed Networks ePDG Administration Guide* | Affirmed Networks evolved Packet Data Gateway and its features, interfaces, and configuration. |
| *Affirmed Networks Gateway Administration Guide* | Affirmed Networks Serving Gateway, Packet Data Network Gateway, Gateway GPRS support node, and Serving GPRS Support Node and associated interfaces and services running on the MCC. This document also describes the gateway features and functions and the command line interface objects and commands used to configure gateway services. Some objects or commands are supported on specific gateways as noted. |
| *Affirmed Networks GTP Proxy Administration Guide* | MCC GPRS Tunneling Protocol (GTP) Proxy, including feature descriptions, interfaces, network deployments, and statistics. |

| Title | Description |
|---|---|
| *Affirmed Networks MCC Feature Overview and Administration Guide for Release 9.0* | Overviews, feature descriptions and configuration information for MCC CUPS, 5G-NSA, ISM/IWM strategies, TAM, and interface enhancements. |
| *Affirmed Networks Overview Guide* | Affirmed Networks MCC, Acuitas Service Management System and the MCC Command Line Interface used to configure, control, and monitor services in the network. This document also includes information on the virtual hardware and software architecture. |
| *Affirmed Networks Pooling License Certificate Generation and Revocation Guide* | Customer portal of the Entitlement Generation System (EGS) to manage entitlements on the Affirmed Networks Pooling Licensing Server. |
| *Affirmed Networks Pooling Licensing Server Installation Guide* | Installation, configuration, operation and maintenance of the Affirmed Networks Pooling Licensing Server (PLS). |
| *Affirmed Networks SNMP MIB and Alarm Reference Guide* | Displaying trap alarm information, the organization of the Affirmed Management Information Base, and information on accessing and reading the Management Information Base. |
| *Affirmed Networks System Administration Guide* | MCC command line interface objects and commands required for the initial system setting, managing user accounts, managing the system, and managing clusters and nodes. This document also provides the command line interface objects and commands for configuring system security and IP protocols. |
| *Affirmed Networks Troubleshooting Guide* | Assists network operators in identifying and troubleshooting common issues associated with Affirmed Networks software. |
| *Affirmed Networks TWAG/TWAP Administration Guide* | MCC Trusted WLAN Access Gateway and the Trusted WLAN AAA Proxy, including feature descriptions, network deployments, interfaces, and statistics. |
| *Affirmed Networks Value-Added Services Administration Guide* | Affirmed Networks Value-Added Services running on the MCC and the MCC command line interface commands and objects used to configure, control, and monitor those services in the network. |
| **Lawful Intercept Module** | |
| *Affirmed Networks Lawful Intercept Module Specification and Provisioning Guide* | Specification for the Affirmed Networks Lawful Intercept Module. This defines the Affirmed Networks X1, X2, and X3 interfaces definition and describes how to provision and manage Lawful Intercept interfaces. The Affirmed Networks Lawful Intercept Module implementation conforms to the standards developed by the European Telecommunications Standards Institute and the United States Communications Assistance for Law Enforcement Act. In this document, the term CALEA applies to both the Lawful Intercept and CALEA interfaces. |
| **Mobile Content Cloud Virtual Platform** | |
| *Mobile Content Cloud Deployment Guide for Dell PowerEdge R630 (SR-IOV Configuration)* | Installation and configuration of the Affirmed Networks Mobile Content Cloud for Dell PowerEdge R630. |

| Title | Description |
|---|---|
| *Mobile Content Cloud Deployment Guide for Dell PowerEdge R740 (SR-IOV Configuration)* | Installation and configuration of the Affirmed Networks Mobile Content Cloud for Dell PowerEdge R740. |
| *Mobile Content Cloud Deployment Guide for HP BladeSystem c7000 (SR-IOV/Non SR-IOV Configuration)* | Installation and configuration of the Affirmed Networks Mobile Content Cloud for HP BladeSystem c7000 (SR-IOV/Non SR-IOV Configuration). |
| *Mobile Content Cloud Deployment Guide for HP ProLiant DL380 Gen9 (SR-IOV/Non SR-IOV Configuration)* | Installation and configuration of the Affirmed Networks Mobile Content Cloud for ProLiant DL380 Gen9 Server (SR-IOV and Non-SR-IOV configurations). |
| *Mobile Content Cloud Deployment Guide for KVM* | Installation and configuration of the Affirmed Networks Mobile Content Cloud for KVM. |
| *Mobile Content Cloud Deployment Guide for RHEL 7.3/7.4 OpenStack Newton Using Packstack (All Interfaces Over SR-IOV)* | Deploying the MCC with SR-IOV over all interfaces using Red Hat Enterprise Linux (RHEL) 7.3/7.4 OpenStack Newton using the Packstack. |
| *Mobile Content Cloud Deployment Guide for RHEL 7.3/7.4 OpenStack Newton Using Packstack (SR-IOV Configuration)* | Deploying the MCC with SR-IOV using Red Hat Enterprise Linux 7.3/7.4 OpenStack Newton using Packstack. |
| *MCC Deployment Guide for RHOPS10 Director – SR-IOV EW/NS Deployments (inc Upgrade of RHOPS10 Director to RHOPS13 Director SR-IOV EW/NS Deployments)* | Deploying the MCC with Red Hat OpenStack Platform 10 (RHOSP10) (OpenStack Newton) Director using SR-IOV EW/NS to RHOSP13/RHEL7.5 (OpenStack Queens) Director using SR-IOV EW/NS. Includes an upgrade guide for RHOPS10 Director to RHOPS13 Director - SR-IOV EW/NS deployments. |
| *MCC Deployment Guide for RHOPS13 Director – SR-IOV EW/NS Deployments* | Deploying the MCC with Red Hat OpenStack Platform 13 (RHOSP13) (OpenStack Queens) using RHOSP Director on the MCC using SR-IOV with HPE/Mellanox 25GB and Dell/Intel Fortville 25 GB NICs. |
| *Mobile Content Cloud Deployment Guide for VMware ESXi* | Deploying the MCC with SR-IOV using Red Hat Enterprise Linux OpenStack Platform. |
| **Mobile Content Cloud Mobility Management Entity (MCC MME)** | |
| *Affirmed Networks MCC C-SGN/MME Deployment Guide for VMware vCloud Director* | Installation of the MME and Gateway product from a platform/solution perspective in the role of C-SGN/MME. This document is specific to the VMware vCloud-based solution. |
| *Affirmed Networks MCC C-SGN/MME Engineering Guide* | Engineering information pertaining to the MCC C-SGN/MME with information related to lab performance tests, engineering/dimensioning aspects of C-SGN/MME and a summary of C-SGN/MME limitations. |
| *Affirmed Networks MCC MME and SGSN Engineering Guide* | LTE MCC MME and the Serving GPRS Support Node engineering information for use by the Affirmed Networks Engineering community. |
| *Affirmed Networks MCC MME and SGSN Engineering Guide - OpenStack Solution* | LTE MCC MME and the Serving GPRS Support Node engineering information that is specific to the OpenStack-based virtualization infrastructure to the Affirmed Networks Engineering community. |

| Title | Description |
|---|---|
| *Affirmed Networks MCC MME and SGSN Engineering Guide – VMware Solution* | LTE MCC MME and the Serving GPRS Support Node engineering information that is specific to the VMware-based virtualization infrastructure to the Affirmed Networks Engineering community. |
| *Affirmed Networks MCC MME CLI Reference Guide* | MCC MME Command Line Interface operational command information that supports configuration, operations, and maintenance of the MCC MME. |
| *Affirmed Networks MCC MME CLI User Guide* | Using the MCC MME Command Line Interface to access, monitor and configure devices on the system. |
| *Affirmed Networks MCC MME Configuration Management Guide* | MCC MME configuration information. |
| *Affirmed Networks MCC MME CSL Reference Guide* | Details regarding Call Summary Logs. The function of Call Summary Logs is to capture events generated by the MME/Serving GPRS Support Node control plane and stream them to a file or an external server. |
| *Affirmed Networks MCC MME Fault Management Reference Guide* | Fault Management information, including descriptions of all MCC MME alarms and logs. |
| *Affirmed Networks MCC MME KVM Solution Integration Guide* | Installation and maintenance of the MME product from a platform/solution perspective performing in the role of MME, SGSN, or a combined node. This document is specific to the KVM-based solution. |
| *Affirmed Networks MCC MME Lawful Interception Interface Guide for Packet Data* | Planning, engineering, administration, or maintenance of the Packet Core in a GPRS, UMTS or LTE network. |
| *Affirmed Networks MCC MME Monitoring Guide* | Monitoring and analysis functions of MCC MME/Serving GPRS Support Node performance management. |
| *Affirmed Networks MCC MME Open Source Acknowledgments* | MCC MME Open Source Software acknowledgments. |
| *Affirmed Networks MCC MME OpenStack Solution Integration Guide* | Information intended for use by support teams and customers who wish to install and maintain the MCC MME product from a platform/solution perspective, performing in the role of MME, Serving GPRS Support Node, or a combined node. This document is specific to the OpenStack-based solution. |
| *Affirmed Networks MCC MME Operator Guide* | Installation and maintenance of the MME product from an application perspective, performing in the role of MCC MME, Serving GPRS Support Node, or a combined node. |
| *Affirmed Networks MCC MME Operator Guide - EPC Solution* | Deploying the MCC MME in an EPC-in-a-box configuration. |
| *Affirmed Networks MCC MME Performance Management Reference Guide* | Performance Management, including a brief description of each Group Set and its associated Group(s). |
| *Affirmed Networks MCC MME Product Overview* | Technical information, functions and capabilities implemented for the virtualized MCC MME node. |

| Title | Description |
|---|---|
| *Affirmed Networks MCC MME REST API Guide* | Managing the local user accounts via REST using the curl commands from a single sign-on system. This includes example commands with various REST operations like GET, POST, PATCH and DELETE. Request and response data can be in XML or JSON format where XML is the default format. |
| *Affirmed Networks MCC MME SGSN Billing Samples* | Call detail record samples and billing information, including selected billing test cases for the MCC MME Serving GPRS Support Node. This document also provides the call detail record that was generated from running the test. |
| *Affirmed Networks MCC MME VMware Solution Integration Guide* | Installation and maintenance of the MME product from a platform/solution perspective performing in the role of MME, Serving GPRS Support Node, or a combined node. This document is specific to the VMware-based solution. |
| *Affirmed Networks MCC MME/SGSN Deployment Guide - Migrating from an Existing MME/SGSN* | Procedures and methods for deploying the MCC MME/SGSN into an existing 2G/3G/4G network and minimizing service interruptions in the process. |
| *Affirmed Networks MCC MME Upgrade Guide* | MME/Serving GPRS Support Node upgrade information. This is a special upgrade procedure that involves a service outage and manual editing of the configuration file. |
| **Virtual Network Function Manager (VNFM)** | |
| *Affirmed Networks VNFM Installation and User's Guide* | Using the Affirmed Virtual Network Function Manager (VNFM) to perform lifecycle management of Virtual Network Functions (VNFs) from Affirmed Networks and third-party vendors. |